

The Program in Action

When you run *etchingdrawer.py*, the output will look like Figure 23-1.

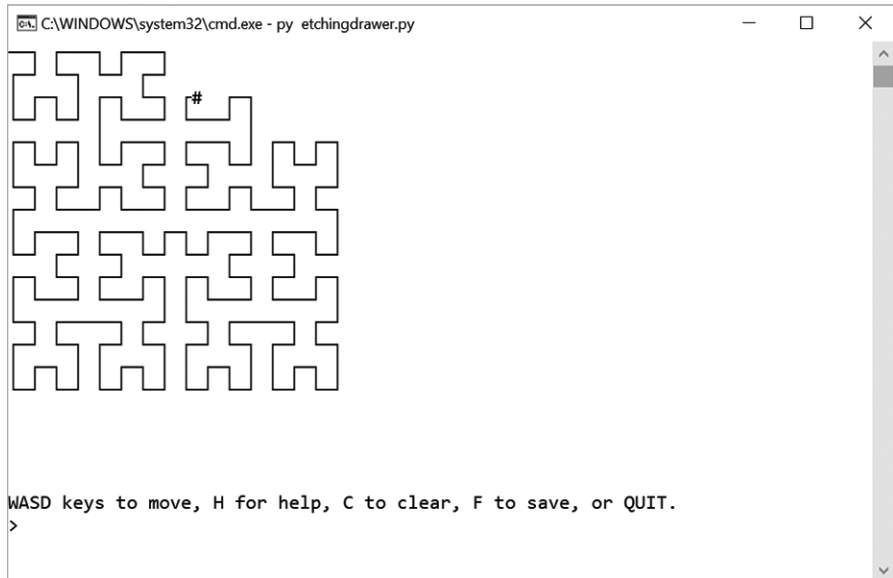


Figure 23-1: A drawing made in the *etching drawer* program

How It Works

Like Project 17, “Dice Math,” this program uses a dictionary stored in a variable named *canvas* to record the lines of the drawing. The keys are (x, y) tuples and the values are strings representing the line shape to draw at those x, y coordinates on the screen. A full list of Unicode characters you can use in your Python programs is given in Appendix B.

Line 126 has a function call to *open()* that passes an *encoding='utf-8'* argument. The reasons are beyond the scope of this book, but this is necessary for Windows to write the line characters to a text file.

1. `"""Etching Drawer, by Al Sweigart al@inventwithpython.com`
2. `An art program that draws a continuous line around the screen using the`
3. `WASD keys. Inspired by Etch A Sketch toys.`
4.
5. `For example, you can draw Hilbert Curve fractal with:`
6. `SDWDDSASDSAASWASSDSASSDWDSDWWAWDDDSASSDWDSDWWAWDWWASAADWAWDDSDW`
7.
8. `Or an even larger Hilbert Curve fractal with:`
9. `DDSAASDDWDDSDWWAWDDDDSDWDDDDSAASDDSAASAAWAAASSDDWDDDDSAASDDSAASAAWA`
10. `ASAAAAWDDWAAASAAWASSDDSAASDDWDDDDSAASDDSAASAAWAAASSDDSAASDDWDDSDWAA`
11. `AWDDDDDDSAASDDWDDSDWAAWDDWAAASAAAWDDWAAWDDDDSDWDDSDWDDDDSAASDDSD`
12. `AAAWAASDDSAASDDWDDSDWAAWDDDDDDSAASDDWDDSDWAAWDDWAAASAAAWDDWA`
13. `AWDDDDSDWAAWDDWAAASAAWASSDDSAASAAWAAASAAAWDDWAAWDDDDSDWAAWAAASAAW`
14. `DWAAWDDDDSDWDDDDSAASDDWDDSDWAAWDD`